Algebraic geometry 2026

Luiz Alberto

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Contents

Chapter 1

Mynotes from 2026 class

Some references used here:

- Mumford's red book [2]
- Humphreys Linear algebraic groups [1]

1.1 Lecture 1 and 2

First some definitions and lemmas worth to remember.

- (Integral extension). Let R be a ring and $S \subset R$ a subring. We say that R is integrally dependent over S if every $r \in R$ is integral over S, that is, there exist a monic polynomial $f \in S[X]$ such that $f(r) = 0$.

  Lemma 1.1.1. Let $S \subset R$ be rings. A finite subset of elements $b_1, \ldots, b_m \in R$ are all integral over S if and only if $S[b_1, \ldots, b_m]$ is finitely generated (as a $S-$module) over S.

  Proof: $(\Rightarrow)$ Let $b \in R$ be integral over S, i.e., there exist

  $$f = X^d + a_1 X^{d-1} + \cdots + a_d \in S[X] \tag{1.1}$$

  such that $f(b) = 0$. Then, using long division, we see that $S[b] = S + \cdots + Sb^{d-1}$ which is finitely generated. Now, if $b_1, \ldots, b_m$ are all integral over S, we can argue by induction that $S[b_1, \ldots, b_{m-1}]$ is finitely generated over S. However, since b_m is integral over $S[b_1, \ldots, b_{m-1}]$, $S[b_1, \ldots, b_m]$ is finitely generated over $[b_1, \ldots, b_{m-1}]$ and it must be finitely generated over S aswell.

  $(\Leftarrow)$ Suppose that $S[b_1, \ldots, b_m] = S\beta_1 + \cdots + S\beta_n$ for some $\beta_i \in R$. Then for any $b \in R$ which fixes $S[b_1, \ldots, b_m]$, we have $b\beta_i = \sum_j a_{ij}\beta_j$ for some $(a_{ij}) \in \mathrm{Mat}_n(S)$. Therefore

  $$0 = (bI - (a_{ij}))^*(bI - (a_{ij}))v = \det(bI - (a_{ij}))v \tag{1.2}$$

where $v = (\beta_1, \ldots, \beta_n)^T$. Therefore $\det(bI - (a_{ij}))\beta_i = 0, \forall i$ and since $1 \in S\beta_1 + \cdots + S\beta_n$, we see that $\det(bI - (a_{ij})) = 0$. Since $\det(X - (a_{ij})) \in S[X]$ is monic, we are done. □

Corollary 1.1.2. (Transitive property of integral dependence). Given rings $R \supset S \supset T$, if R is integral over S and S is integral over T, then R is integral over T.

Question: Do we need Noetherian hypothesis using this statement?

Proof: Let $r \in R$, then by hypothesis there exist $f = X^d + s_1 X^{d-1} + \cdots \in S[X]$ such that $f(r) = 0$. Define $A = T[s_1, \ldots, s_d]$, then $A[r]$ is a finitely generated $A-$module because $S \subset A$ and lemma (1.1.1). Now, since A is a finitely generated $T-$module by lemma (1.1.1), we see that $A[r]$ is a finitely generated $T-$module. Now, assuming T is Noetherian we see that $T[r]$ is finitely generated as a $T-$module. □

Remark 1.1.3. We can certainly say (using the previous proof) that if $r \in R$, then $S[r]$ is finitely generated as a $T-$module.

Lemma 1.1.4. Let $K \supset k$ (k a perfect field) be a finitely generated extension field with transcendence degree $\partial_k(K) = d$, then there exist generators $z_1, \ldots, z_d, z_{d+1}$ such that

(1) $K = k(z_1, \ldots, z_{d+1})$
(2) $\{z_1, \ldots, z_d\}$ are algebraicaly independent
(3) z_{d+1} is separable over $k(z_1, \ldots, z_d)$

Lemma 1.1.5. (Noether's normalization lemma - Mumford's red book). Let R be a finitely generated algebra over k. Assume also that R is an integral domain. If R has transcende degree n over k (which means that $\partial_k \mathrm{Frac}(R) = n$), then there exist $x_1, \ldots, x_n \in R$ algebraicaly independent and such that R is integral over $k[x_1, \ldots, x_n]$.

TODO: Study prove this version of the lemma using the statement made in class.

Lemma 1.1.6. (Noether's normalization lemma - Statement made in class). Let R be a finitely generated $k-$algebra. Then there exist $x_1, \ldots, x_n \in R$ algebraically independent over k such that R is integral over $k[x_1, \ldots, x_n]$.

Proof (Nagata): Since R is finitely generated as a k-algebra, there exist generators $y_1, \ldots, y_m$ such that $R = k[y_1, \ldots, y_m]$. Now suppose that $y_1, \ldots, y_m$ are algebraically independent, then we can take $x_i = y_i$. If not, then there exist $p \in k[Y_1, \ldots, Y_m] \setminus 0$ such that $p(y_1, \ldots, y_m) = 0$. Write

$$p(Y_1, \ldots, Y_m) = \sum_{\alpha \in \mathbb{Z}^m_{\geq} 0} a_\alpha Y_1^{\alpha_1} \cdots Y_m^{\alpha_m} \tag{1.3}$$

and define a new polynomial

$$\begin{aligned} q(Y_1, \dots, Y_m) &= p(Y_1, Y_2 + Y_1^{r_2}, \dots, Y_m + Y_1^{r_m}) \\ &= \sum_{\alpha} a_\alpha (Y_1^{\alpha_1 + r_2\alpha_2 + \cdots + r_m\alpha_m} + \cdots) \end{aligned} \tag{1.4}$$

for some choice of positive integers $r_2, \dots, r_m$. Take a positive number $b > \alpha_i$, $\forall \alpha_i$ such that $a_\alpha \neq 0$. If we take $r_i = b^i$, then $\alpha_1 + \alpha_2 b + \cdots + \alpha_m b^{m-1}$ are all different numbers in base b. So, from different powers of the form $\alpha_1 + r_2\alpha_2 + \cdots + r_m\alpha_m$ we have a maximal one $d = \max\{\alpha_1 + \cdots r_m\alpha_m : a_\alpha \neq 0\}$ and

$$q(Y_1, \dots, Y_m) = aY_1^d + \text{smaller degree terms} \in k[Y_2, \dots, Y_m][Y_1], \ a \neq 0. \tag{1.5}$$

Now, if we let $z_i = y_i - y_1^{r_i} \in R, i = 2, \dots, m$, we see that $q(y_1, z_2, \dots, z_m) = p(y_1, \dots, y_m) = 0$ and y_1 is integral over $k[z_2, \dots, z_m]$. By induction on number of generators, there exist $x_1, \dots, x_n$ algebraically independent such that $k[z_2, \dots, z_m]$ is integral over $k[x_1, \dots, x_n]$. So, by lemma 1.1.1 and its corollary (1.1.1), we see that $k[y_1, z_2, \dots, z_m]$ is integral over $k[x_1, \dots, x_m]$. Now, by definition, it's clear that $k[y_1, z_2, \dots, z_m]$ is a finitely generated $R-$module. So by the corollary (1.1.2) again, we see that R is integral over $k[x_1, \dots, x_m]$. $\square$

Lemma 1.1.7. Let $S \subset R$ be a ring extension. If R is a field, which integral over S, then S is a field.

Proof: Let $s \in S \setminus 0$ and $r \in R$ be its inverse. Since R is integral over S, there exist $f = X^d + a_1X^{d-1} + \cdots \in S[X]$ be such that $f(r) = 0$. Therefore

$$0 = s^{d-1}f(r) = r + a_1 + sa_2 + \cdots + s^{d-1}a_0 \Rightarrow r \in S. \tag{1.6}$$

$\square$

Remark 1.1.8. Let $L|K$ be a field extension. Then

$$\begin{tikzcd} L|K \text{ finite extension} \arrow[r, bend left] & L|K \text{ algebraic extension} \arrow[l, bend left, "+\text{ finitely generated extension}"] \end{tikzcd}$$

Theorem 1.1.9. (Zariski). Let k be a field and $\mathfrak{m} \subset R = k[X_1, \dots, X_m]$ be a maximal ideal. Then $L = R/\mathfrak{m}$ is a finite extension field of k.

Proof: By Noether normalization lemma (1.1.6), there exist $a_1, \dots, a_n \in L$ algebraically independent such that L is integral over $k[a_1, \dots, a_n]$. By lemma (1.1.7), we conclude that $k[a_1, \dots, a_n]$ is a field. Suppose that $n > 0$, so a_1 has an inverse. This means that $a_1 g(a_1, \dots, a_n) = 1$ for some $g \in k[X_1, \dots, X_n]$, which constradicts the fact that $a_1, \dots, a_n$ are alegrbaically independent. Then we conclude that L is an integral (and then algebraic) over k. However, since L is a finitely generated extension ($L = k(\overline{X}_1, \dots, \overline{X}_m)$) over k we conclude that L is in fact a finite field extension over k. $\square$

Theorem 1.1.10. Suppose that k is an algebraically closed field. Let $I \subset R = k[X_1, \dots, X_m]$ be an ideal, then $V(I) = \varnothing \iff I = k[X_1, \dots, X_m]$.

Proof: If $I \neq R$, then there exist a maximal ideal $\mathfrak{m} \supset I$. So $V(I) \supset V(\mathfrak{m})$ and it's sufficient to prove that $V(\mathfrak{m}) \neq \varnothing$. By theorem (1.1.9), we have that $L = R/\mathfrak{m}$ is a finite $k-$extension, so we must have $L = k$ because k is algebraically closed. So there exist $a_i \in k, i = 1, \dots, m$ such that $X_i - a_i \in \mathfrak{m}$ and $(X_1 - a_1, \dots, X_m - a_m) \subset \mathfrak{m}$. However $\varphi : R \to k : F \mapsto F(a_1, \dots, a_m)$ has kernel $\mathrm{Ker}(\varphi) = (X_1 - a_1, \dots, X_m - a_m)$ and since k is a field, $R/\mathrm{Ker}(\varphi)$ must be a field and $\mathrm{Ker}(\varphi)$ is a maximal ideal. Therefore we conclude that $\mathfrak{m} = (X_1 - a_1, \dots, X_m - a_m)$. Finally $V(X_1 - a_1, \dots, X_m - a_m) = \{a_1, \dots, a_m\} \neq \varnothing$. $\square$

Remark 1.1.11. The hypothesis $k = \overline{k}$ in the previous theorem is crucial. For example if $k = \mathbb{R}$, we have $V(x^2 + 1) = \varnothing$.

Corollary 1.1.12. From the proof of the previous theorem, we see that there exist a bijection

$$\begin{aligned} k^n &\longleftrightarrow \left\{ \begin{matrix} \text{maximal ideals} \\ \mathfrak{m} \subset R \end{matrix} \right\} \\ (a_1, \dots, a_n) &\longmapsto (X_1 - a_1, \dots, X_n - a_n) \end{aligned} \tag{1.7}$$

Theorem 1.1.13. (Hilbert Nullstelensatz). Let $A \subset k[X_1, \dots, X_n]$ be an ideal, then

$$IV(A) = \sqrt{A}. \tag{1.8}$$

Proof: The inclusion $\sqrt{A} \subset IV(A)$ is easy. Now, assume $g \in IV(A)$, where $A = (f_1, \dots, f_m)$. This means that we have the hypothesis:

$$f_i(a_1, \dots, a_n) = 0,\ \forall i = 1, \dots, m \Rightarrow g(a_1, \dots, a_n) = 0\ (*) \tag{1.9}$$

Now, define the ideal

$$I = Ak[X_1, \dots, X_{n+1}] + (1 - g(X_1, \dots, X_n)X_{n+1}). \tag{1.10}$$

If I is a proper ideal, then by corollary (1.1.12), we have $I \subset (X_1 - a_1, \dots, X_{n+1} - a_{n+1})$, which is the kernel of the evaluation map $k[X_1, \dots, X_{n+1}] \to k : f \mapsto f(a_1, \dots, a_{n+1})$. This means that

$$\begin{cases} f_i(a_1, \dots, a_n) = 0,\ \forall i = 1, \dots, m \\ g(a_1, \dots, a_n)a_{n+1} = 1 \end{cases} \tag{1.11}$$

which contradicts our initial hypothesis $(*)$. Therefore, I is not a proper ideal, so $1 \in I$ and there exists $h_1, \ldots, h_{m+1} \in k[X_1, \ldots, X_{n+1}]$ such that

$$1 = \sum_{j=1}^{m} h_j(X_1, \ldots, X_{n+1}) f_j(X_1, \ldots, X_n) + h_{m+1}(X_1, \ldots, X_{n+1})(1 - g(X_1, \ldots, X_n)X_{n+1}). \tag{1.12}$$

Now, by projecting this equation onto $k[X_1, \ldots, X_{n+1}]/(1 - gX_{n+1})$, we get

$$g^{\ell}(X_1, \ldots, X_n) = \sum_{j=1}^{m} h_j(X_1, \ldots, X_{n+1}) g^{\ell} \times f_j(X_1, \ldots, X_n) \tag{1.13}$$

modulo $(1 - gX_{n+1})$, where $H_j = h_j(X_1, \ldots, X_{n+1})g^{\ell} \in k[X_1, \ldots, X_n]$, $\forall j$ if $\ell \gg 0$. Since $\{\overline{X}_1, \ldots, \overline{X}_{n+1}\}$ is algebraically free in $k[X_1, \ldots, X_{n+1}]/(1 - gX_{n+1})$, we conclude that, in fact, we have equality

$$g^{\ell}(X_1, \ldots, X_n) = \sum_{j=1}^{m} H_j(X_1, \ldots, X_n) \times f_j(X_1, \ldots, X_n) \tag{1.14}$$

in $k[X_1, \ldots, X_n]$. $\square$

Remark 1.1.14. We could have used the homomorphism

$$k[X_1, \ldots, X_{n+1}] \to k(X_1, \ldots, X_n) : X_{n+1} \mapsto g,\ X_i \mapsto X_i,\ i \neq n+1 \tag{1.15}$$

at the end of the previous proof.

1.2 Lecture 3

Definition 1.2.1. We define an affine algebraic set $V \subset k^n$ to be a set of the form $V = V(S)$ for some $S \subset k[\underline{X}]$.

Notation 1.2.2. Let V be an affine algebraic set. We write

$$D_V(g) = \{p \in V : g(p) \neq 0\} = V \setminus V(g). \tag{1.16}$$

These open sets are called "principal open subsets".

Lemma 1.2.3. Any open subset $U \subset V$ of an affine algebraic set is quasicompact.

Proof: Consider a covering $U = \bigcup_{\lambda \in \Lambda} U_\lambda$. Observe that $V \setminus U = V(g_1, \ldots, g_m)$, so we have the equality $U = \bigcup_{i=1}^{m} D_V(g_i)$. In particular, we see that principal open subsets form a basis for the Zariski topology on V. Therefore, it's sufficient to prove that a principal open subset is quasicompact. So we assume $U = D_V(g)$. Furthermore, we have $U_\lambda = \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i})$ and

$$U = \bigcup_{\lambda\in\Lambda} \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i}). \tag{1.17}$$

So, without lost of generalizaiton, we can assume $U = \bigcup_{\lambda\in\Lambda} D_V(g_\lambda)$ from beginning. Now

$$\begin{aligned} D_V(g) = \bigcup_{\lambda\in\Lambda} D_V(g_\lambda) &\iff \\ V \cap V(g) = V \cap V(\langle g_\lambda : \lambda \in \Lambda\rangle) &\iff \\ V \cap V(g) = V \cap V(\langle g_\lambda : \lambda \in I\rangle) \ (I \subset \Lambda \ \text{ finite}). \end{aligned} \tag{1.18}$$

In the last equation we used Hilbert basis theorem: The sequence

$$\langle g_{\lambda_1}\rangle \subset \langle g_{\lambda_1}, g_{\lambda_2}\rangle \subset \cdots \tag{1.19}$$

must terminate at some point, because $k[\underline{X}]$ is Noetherian. Then we conclude that $D_V(g) = \bigcup_{\lambda\in I} D_V(g_\lambda)$. □

Lemma 1.2.4. Suppose that k is an infinity field, then open sets $U \subset k^n$ are dense in the Zariski topology.

Proof: Since basic open sets form a basis for the Zariski topology on k^n, it's sufficient to prove that $D(f)$ is dense in k^n. Let $\overline{D(f)} = V(g_1, \dots, g_m)$, $m \geq 1$ and $x \in k^n$, then we have for all $i = 1, \dots, m$:

$$f(x) \neq 0 \Rightarrow g_i(x) = 0 \Rightarrow fg_i(x) = 0, \ \forall x \in k^n. \tag{1.20}$$

Therefore, since k is infinity, we have $fg_i = 0 \in k[\underline{X}]$, $\forall i$ and $g_i = 0$, $\forall i$. Then we see that $\overline{D(f)} = k^n$. □

Definition 1.2.5. (Regular functions). Let $V \subset k^n$ be an affine algebraic set and $U \subset V$ be an open set. A map $\varphi : U \to k$ is said to be *regular* if for any $p \in U$, there exist $p \in U' \subset U$, polynomials $f, g \in k[\underline{X}]$ such that $g(q) \neq 0$, $\forall q \in U'$ and $\varphi|_{U'} = f/g|_{U'}$.

Notation 1.2.6. Let V be an affine algebraic set and let $U \subset V$ be an open set, we denote $\mathcal{O}_V(U) = \{U \overset{\text{regular}}{\to} k\}$.

Lemma 1.2.7. Let $V \subset k^n$ be an affine algebraic set and $g \in k[X_1, \ldots, X_n]$. Then the ser of regular functions $\varphi : D_V(g) \to k$ is of the form f/g^m for some $f \in k[X_1, \ldots, X_n]$ and $m > 0$. So

$$\mathcal{O}_V(D_V(g)) \cong \widetilde{\psi}(k[X_1, \ldots, X_n]_g) \tag{1.21}$$

where $\psi : k[X_1, \ldots, X_n] \to \mathcal{O}_V(D_V(g))$ is the map that takes a polynomial and restricts it to $D_V(g)$ and $\widetilde{\psi}$ is the corresponding map induced by the universal property of localization.

Proof: First we have $D_V(g) = \bigcup_{i=1}^{\ell} D_V(g_i)$ and since φ is regular, there exists $f_i, h_i \in k[\underline{X}]$, $i = 1, \ldots, \ell$ such that

$$q \in D_V(g_i) \Rightarrow h_i(q) \neq 0 \text{ (so } q \in D_V(h_i)),\ \varphi|_{D_V(g_i)} = \frac{f_i}{h_i}\bigg|_{D_V(g_i)}. \tag{1.22}$$

Now, we have two fundamental relations:

$$\begin{cases} D_V(g_i) \subset D_V(h_i),\ \forall i = 1, \ldots, \ell \\ D_V(g) = \displaystyle\bigcup_{i=1}^{\ell} D_V(g_i) \end{cases} \tag{1.23}$$

Using $V = V(F_1, \ldots, F_m)$ and Hilbert Nullstelensatz (1.1.13), we get

$$\begin{cases} \sqrt{(F_1, \ldots, F_m, g_i)} \subset \sqrt{(F_1, \ldots, F_m, h_i)},\ \forall i = 1, \ldots, \ell \quad (*) \\ \sqrt{(F_1, \ldots, F_m, g)} = \sqrt{(F_1, \ldots, F_m, g_1, \ldots, g_\ell)} \quad (**) \end{cases} \tag{1.24}$$

From $(*)$, we deduce that $g_i^{n_i} = \sum_j A_j F_j + B_i h_i$ and restricting to V we get $g_i^{n_i} = B_i h_i$. Now, since $\varphi|_{D_V(g_i)} = \frac{B_i f_i}{g_i^{n_i}}$ and $D_V(g_i) = D_V(g_i^{n_i})$, we can change notation $B_i f_i \to f_i$ and $g_i^{n_i} \to g_i$.

Now we do a trick: since $g_j f_i - f_j g_i = 0$ on $D_V(g_i) \cap D_V(g_j) = D_V(g_i g_j)$, we get

$$g_i g_j (g_j f_i - f_j g_i) = 0 \text{ on } V. \tag{1.25}$$

Furthermore, we still have $\varphi|_{D_V(g_i)} = g_i f_i / g_i^2$. So we change notation again: $g_i^2 \to g_i$ and $g_i f_i \to f_i$, so we have

$$g_i f_j - g_j f_i = 0,\ \forall i, j = 1, \ldots, \ell \quad (***). \tag{1.26}$$

Now, by equation $(**)$, there exist $m > 0$ such that

$$g^m = \sum_j A_j F_j + \sum_k B_k g_k \tag{1.27}$$

and we get on V:

$$g^m = \sum_k B_k g_k. \tag{1.28}$$

Now, by using equation $(***)$, we have

$$f_i g^m = \sum_k B_k f_i g_k = \sum_k B_k f_k g_i = (\sum_k B_k f_k) g_i. \tag{1.29}$$

So if $q \in D_V(g_i)$, we have $\varphi(q) = \frac{f_i}{g_i}(q) = \frac{f}{g^m}(q)$, where we set $f = \sum_k B_k f_k$. $\square$

Corollary 1.2.8. Since $V = D_V(1)$, we have

$$\mathcal{O}_V(V) = \psi(k[X_1, \ldots, X_n]) \cong k[\underline{X}]/I(V) = k[V]. \tag{1.30}$$

where $\psi : k[\underline{X}] \to O_V(V)$ is as in the previous theorem.

Definition 1.2.9. A morphism between two affine algebraic sets $V_1 \subset k^n$ and $V_2 \subset k^m$ is a function $\varphi : V_1 \to V_2$ such that for any open set $U \subset V_2$, we have

$$\varphi_U^{\#} : \mathcal{F}(U, k) \overset{-\circ\varphi}{\to} \mathcal{F}(\varphi^{-1}(U), k),\ \varphi_U^{\#}(\mathcal{O}_{V_1}(U)) \subset \mathcal{O}_{V_2}(\varphi^{-1}(U)). \tag{1.31}$$

Using more fancing words, the previous definition says that a continuous map $\varphi : V_1 \to V_2$ is a morphism if the composition map $\varphi_U^{\#} : \mathcal{F}(U, k) \overset{-\circ\varphi}{\to} \mathcal{F}(\varphi^{-1}(U), k)$ induces a sheaf morphism

$$\varphi^{\#}(U) : \mathcal{O}_{V_2} \to f_* \mathcal{O}_{V_1}. \tag{1.32}$$

We use the notation $f_*\mathcal{O}_{V_1}$ for direct image sheaf, it's defined by $f_*\mathcal{O}_{V_1}(U) = \mathcal{O}_{V_1}(f^{-1}(U))$.

1.3 Lecture 4

Definition 1.3.1. A <u>function space</u> is a topological space X with a sheaf $\mathcal{O}_X$ such that for any open set $U \subset X$, we have

(1) $\mathcal{O}_X(U) \overset{\text{subring}}{\subset} \{U \overset{\text{continuous}}{\to} k\}$

(2) Let $f \in \mathcal{O}_X(U)$, then f has an inverse in $\mathcal{O}_X(U)$ if and only if $f(q) \neq 0$, $\forall q \in U$.

Lemma 1.3.2. An affine algebraic set $V \subset k^n$ equiped if the sheaf of regular functions is a function space.

Proof: Let $U \subset X$ be an open set and $f \in \mathcal{O}_X(U)$. Then, by definition, there exist an open cover $U = \bigcup_\lambda U_\lambda$ such that $f|_{U_\lambda} = f_\lambda / g_\lambda$, where $g_\lambda(q) \neq 0 \; \forall q \in U_\lambda$. To prove that f is continuous, it's sufficient to prove that $f^{-1}(a) \subset V$ is closed for all $a \in k$. So we prove that $(f|_{U_\lambda})^{-1}(a)$ is closed in V for all λ, which can be verified by the equality

$$V \cap V(f_\lambda - a g_\lambda) = (f|_{U_\lambda})^{-1}(a). \tag{1.33}$$

The second item is clear. □

Definition 1.3.3. Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be two function spaces. We define a morphism $X \to Y$ to be a a continuous map $f : X \to Y$ which induces a sheaf map $f^{\#} : \mathcal{O}_Y \to f_* \mathcal{O}_Y$ in the sence that if $f^{\#}(U)(g) = g \circ f$ for $g \in \mathcal{O}_Y(U)$, then $f^{\#}(U)(\mathcal{O}_Y(U)) \subset \mathcal{O}_X(f^{-1}(U))$.

Definition 1.3.4. An affine algebraic variety is a function space $(X, \mathcal{O}_X)$ isomorphic to an affine algebraic set with Zariski topology and sheaf of regular functions.

Definition 1.3.5. Let $(X, \mathcal{O}_X)$ be a function space, and $U \subset X$ be an open set with induced topology and $\mathcal{O}_U = \mathcal{O}_X|_U$. Then $(U, \mathcal{O}_U)$ is a function space called induced open subspace.

Definition 1.3.6. An abstract algebraic variety is a function space $(X, \mathcal{O}_X)$ such that there exist an open covering $X = \bigcup_\lambda U_\lambda$ with the property that for each λ, the induced open subspace $(U_\lambda, \mathcal{O}_{U_\lambda})$ is an affine algebraic variety.

Remark 1.3.7. In many texts we have the additional hypothesis of irredutibility in the definition of algebraic varieties.

Definition 1.3.8. An affine algebraic variety is said to be quasi-affine if it's isomorphic, as a function space, to an induced open subspace of an affine algebraic variety.

Lemma 1.3.9. Let $(X, \mathcal{O}_X)$ be a function space and $Y \subset k^n$ be a quasi affine algebraic set

Lemma 1.3.10. Let $(X, \mathcal{O}_X)$ be a function space and $Y \subset k^n$ be a quasi-affine algebraic set. Then $f : X \to Y$ is a morphism if and only $f = (g_1, \ldots, g_n)$ with $g_i \in \mathcal{O}_X(X)$. Here, we think Y as an induced open subspace of an affine algebraic set.

Proof: $(\Rightarrow)$ The coordinate functions $x_i|_Y : Y \to k$ are regular functions, by definition. Now $g_i = x_i \circ f = f^{\#}(Y)(x_i) \in \mathcal{O}_X(f^{-1}(Y)) = \mathcal{O}_X(X)$, because $x_i \in \mathcal{O}_Y(Y) = \{Y \overset{\text{regular}}{\to} k\}$.

$(\Leftarrow)$ A closed set of Y is of the form $V(F) \cap Y$ (would not be of the form $V(F_1, \ldots, F_m)$ in general ?). However $f^{-1}(V(F) \cap Y) = \{p \in X : F(g_1, \ldots, g_n)(p) = 0\}$ which is closed in X provided we show that $F(g_1, \ldots, g_n)$ is continuous. However, since $\mathcal{O}_X(X)$ is a ring, we have $F(g_1, \ldots, g_n) \in \mathcal{O}_X(X)$. Furthermore, if $U \subset Y$ is an open set and $\varphi : U \to k$ is regular, we need to show that $f^{\#}(U)(\varphi) \in \mathcal{O}_X(f^{-1}(U))$. Let $U' \subset U$ be such that $\varphi|_{U'} = F/G,\ G(q) \neq 0\ \forall q \in U'$. Then we have

$$f^{\#}(U')(\varphi) = \varphi \circ f|_{f^{-1}(U')} = \left.\frac{F(g_1, \ldots, g_n)}{G(g_1, \ldots, g_n)}\right|_{f^{-1}(U')}. \tag{1.34}$$

Furthermore

$$G \circ f(q) = G(g_1, \ldots, g_n)(q) \neq 0,\ \forall q \in f^{-1}(U'). \tag{1.35}$$

Since $G(g_1, \ldots, g_n)|_{f^{-1}(U')} \in \mathcal{O}_X(f^{-1}(U'))$, the previous equation shows that the inverse $1/G(g_1, \ldots, g_n)$ also is an element of $\mathcal{O}_X(f^{-1}(U'))$. Now, let $U = \bigcup_\lambda U_\lambda,\ \varphi|_{U_\lambda} = F_\lambda/G_\lambda$, then we see that

$$f^{\#}(U_\lambda)(\varphi|_{U_\lambda}) \subset \mathcal{O}_X(f^{-1}(U_\lambda)). \tag{1.36}$$

The glueing axiom shows that $f^{\#}(U)(\varphi) \in \mathcal{O}_X(f^{-1}(U))$. □

Lemma 1.3.11. Let $V \subset k^n$ be an affine algebraic set and $D_V(F) \subset V$ an affine algebraic set. Then the induced open subspace $(D_V(F), \mathcal{O}_{D_V(F)})$ is a quasi-affine algebraic variety, by definition. However, we have more, it's also an affine algebraic variety.

Proof: Let $V = V(G_1, \ldots, G_m)$ and define the map

$$\varphi : D_V(F) \to V(x_0F - 1, G_1, \ldots, G_m) \subset k^{n+1} : (a_1, \ldots, a_n) \mapsto (\frac{1}{F(a_1, \ldots, a_n)}, a_1, \ldots, a_n) \tag{1.37}$$

By the lemma (1.3.10), we see that φ is a morphism. Furthermore, it has a set theoretical inverse given by the projection

$$\psi : V(x_0F - 1, G_1, \ldots, G_m) \to D_V(F) : (a_0, \ldots, a_n) \mapsto (a_1, \ldots, a_n). \tag{1.38}$$

By the same lemma, this map is also a morphism. Since $f\psi = \mathbb{1}_{D_V(x0F-1,\ldots)}$ and $\psi f = \mathbb{1}_{D_V(F)}$, we see that φ is an isomorphism (the induced sheaf morphism would be an isomorphism by construction). □

Remark 1.3.12. Let $(X, \mathcal{O}_X)$ be an affine algebraic variety and $\varphi : (V, \mathcal{O}_V) \to (X, \mathcal{O}_{\mathcal{X}})$ an isomorphism, where $v \subset k^n$ is an algebraic set. Then this isomorphism, induces another isomorphism

$$(D_V(F), \mathcal{O}_{D_V(F)}) \to (X_f, \mathcal{O}_{X_f}) \tag{1.39}$$

where $F = f \circ \varphi$ and $X_f = \varphi(D_V(F)) = \{p \in X : f(p) \neq 0\}$.

Corollary 1.3.13. A quasi-affine algebraic variety is an abstract algebraic variety.

Proof: A quasi-affine algebraic variety has a topological basis consisting of principal open subsets, which are affine algebraic varieties.

1.4 Lecture 5

Chapter 2

Algebraic Groups

2.1 Basic definitions

Some important observations:

- If $\mathbb{F}$ is a finite field, the map which takes a polynomial in $\mathbb{F}[X_1,\ldots,X_n]$ and send to a function $\mathbb{F}^n \to \mathbb{F}$ is not an isomorphism. For example, if $\mathbb{F}$ has q elements, then $\prod_{q\in\mathbb{F}}(X-q) \in \mathbb{F}[X] \mapsto 0$. However, if $\mathbb{F}$ is infinity and $a_nX^n + a_{n-1}X^{n-1} + \cdots + a_0 = 0$, then we can select distinc elements $\alpha_0,\ldots,\alpha_n$ and we conclude that

$$\begin{pmatrix} a_0 & \ldots & a_n \end{pmatrix} \begin{pmatrix} 1 & 1 & \cdots & 1 \\ \alpha_0 & \alpha_1 & \cdots & \alpha_n \\ \vdots & \vdots & \cdots & \vdots \\ \alpha_0^n & \alpha_1^n & \cdots & \alpha_n^n \end{pmatrix} = 0. \tag{2.1}$$

  However $\det(\alpha_i^j)_{i,j=0}^n = \prod_{i<j}(\alpha_i - \alpha_j) \neq 0$ and we conclude that $a_0,\ldots,a_n = 0$. Now, if $f \in \mathbb{F}[X_1,\ldots,X_n]$, then we can write (assumin X_1 appears in f):

$$f = g_d(X_2,\ldots,X_n)X_1^d + \cdots + g_1(X_2,\ldots,X_n)X_1 + g_0(X_2,\ldots,X_n). \tag{2.2}$$

  So the equality $f = 0$, the previous step for the case $\mathbb{F}[X]$ and an induction on the number of variables yields $g_d,\ldots,g_0 = 0$. Then we see that if $|\mathbb{F}| = \infty$, we can identify the ring $\mathbb{F}[X_1,\ldots,X_n]$ with polynomial functions $\mathbb{F}^n \to \mathbb{F}$.

  As a consequence of this story, we see that if $|\mathbb{F}| = \infty$, then

$$I(\mathbb{F}^n) = (0) \subset \mathbb{F}[X_1,\ldots,X_n]. \tag{2.3}$$

- (Hilbert basis theorem). The Hilbert basis theorem states that if A is noetherian, then $A[X]$ is noetherian. Since $\mathbb{F}[X]$ is a PID (and in consequence noetherian), we conclude by an induction on the number of variables that $\mathbb{F}[X_1,\ldots,X_n]$ is noetherian.

Remark 2.1.1. A ring is said to be noetherian if it satisfy the ascending chain condition, or equivalently, if all of its ideals are finitely generated.

- (Algebraic sets). Let $S \subset \mathbb{F}[X_1, \ldots, X_n]$, then $V(S) = \{a \in \mathbb{F}^n : f(a) = 0\}$. It's immediate that $V(S) = V((S))$. The relation $V(\bigcup_{i\in I} S_i) = \bigcap_{i\in I} V(S_i)$ allow us to define the Zariski topology on $\mathbb{F}^n$. Moreover, for any subset $V \subset \mathbb{F}^n$, we let $I(V) = \{f \in \mathbb{F}[X_1, \ldots, X_n] : f(a) = 0, \ \forall a \in V\}$. Then we have

$$\begin{cases} VI(W) = \overline{W} \\ IV(\mathfrak{a}) = \sqrt{\mathfrak{a}} \end{cases} \tag{2.4}$$

We also have the important relation

$$V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b}). \tag{2.5}$$

In fact

$$V(\mathfrak{a}) \cup V(\mathfrak{b}) \underset{\mathfrak{a}\cap\mathfrak{b}\subset\mathfrak{a},\mathfrak{b}}{\subset} V(\mathfrak{a} \cap \mathfrak{b}) \underset{\mathfrak{a}\mathfrak{b}\subset\mathfrak{a}\cap\mathfrak{b}}{\subset} V(\mathfrak{a}\mathfrak{b}) = V(\{fg : f \in \mathfrak{a}, g \in \mathfrak{b}\}) \subset V(\mathfrak{a}) \cup V(\mathfrak{b}). \tag{2.6}$$

- (Coordinate rings). Let $\mathfrak{a} = I(V)$ where V is an algebraic set. We define the coordinate ring $A[V] = \mathbb{F}[X_1, \ldots, X_n]/\mathfrak{a}$.

- (LEX order). We define an ordering on $\mathbb{Z}^n_{\geq 0}$ by $\alpha > \beta$ if for the smallest i such that $\alpha_i \neq \beta_i$ we have $\alpha_i > \beta_i$. So, with a fixed LEX on $\mathbb{F}[X_1, \ldots, X_n]$, we can definte the leading term of a polynomial: $\mathrm{LT}(f) = a_\alpha X^\alpha$, where α is the smallest tuple, relative to LEX such that $a_\alpha \neq 0$. In the case $\mathrm{LT}(f) = a_\alpha X^\alpha$, we also write $\mathrm{LM}(f) = X^\alpha$.

- (Groebner Basis). Let $\mathfrak{a}$ be an ideal in $\mathbb{F}[X_1, \ldots, X_n]$. A groebner basis of $\mathfrak{a}$ is a finite subset $G \subset \mathfrak{a}$ such that $(\mathrm{LT}(g) : g \in G) = \mathrm{LT}(\mathfrak{a})$.

Example 2.1.2. Let $R = \mathbb{F}[X_1, \ldots, X_n, Y_1, \ldots, Y_n]$ with LEX: $X_1 > \cdots > X_n > Y_1 > \cdots > Y_n$ and define an ideal $\mathfrak{a} = (X_1 - g_1, \ldots, X_n - g_n)$ for $g_i \in \mathbb{F}[Y_1, \ldots, Y_n]$. Since $\mathrm{LT}(X_i - g_i) = X_i$, it's clear that $(X_1, \ldots, X_n) \subset \mathrm{LT}(I)$. Now, if $f \in R$ and $X_j \nmid \mathrm{LT}(f)$, $\forall j$, then $f \in \mathbb{F}[Y_1, \ldots, Y_n]$ because its smallest term has power $\alpha \in \{0\} \times \mathbb{Z}^n$. So, we need to show that $I \cap \mathbb{F}[Y_1, \ldots, Y_n] = (0)$. This can be seen by proving that the kernel of $\varphi : R \to \mathbb{F}[Y] : X_i \mapsto g_i, Y_i \mapsto Y_i$ is I. The only nontrivial inclusion is $\mathrm{Ker}(\varphi) \subset I$. To see this use a division algorithm to write

$$f = h_1(X_1 - g_1) + \cdots + h_n(X_n - g_n) + r \tag{2.7}$$

where $r \neq 0$ implies that no term of r is divisible by $\mathrm{LT}(X_i - g_i) = X_i, \ \forall i$. By applying φ on both sides of this equation we have the desired inclusion. From this we see that $G = (X_1 - g_1, \ldots, X_n - g_n)$ is a groebner basis for I.

- (Hilbert Polynomial).

Definition 2.1.3. Let $I \subset \mathbb{F}[X_1, \ldots, X_n]$ be an ideal. We deffine the affine Hilbert function:

$$^a\mathrm{HF}_I : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0} : s \mapsto \dim_{\mathbb{F}}(\frac{\mathbb{F}[X_1, \ldots, X_n]_{\leq s}}{I_{\leq s}}). \tag{2.8}$$

Example 2.1.4. For example, the number $^a\mathrm{HF}_{(0)}(s)$ is the number of n-tuples of nonnegative integers whose sum is $\leq s$. To compute this, write the generating function

$$\prod_{i=1}^{n} \frac{1}{1 - X_i} = \sum_{\alpha \in \mathbb{Z}^n_{\geq 0}} X^\alpha. \tag{2.9}$$

Now put $X_i = t$, $\forall i$. We see that

$$(1-t)^{-n} = \sum_{s \geq 0} \binom{-n}{s} (-1)^s t^s = \sum_{s \geq 0} \binom{n+s}{s} t^s. \tag{2.10}$$

So $^a\mathrm{HF}_{(0)}(s) = \binom{n+s}{s} = \frac{1}{n!} s^n + \cdots$.

Example 2.1.5. Let $f \in \mathbb{F}[X_1, \ldots, X_n]$ be a degree d polynomial. Then

$$^a\mathrm{HF}_{(f)}(s) = \binom{s+n}{s} - \binom{s-d+n}{s-d} = \frac{d}{(n-1)!} s^{n-1} + \cdots . \tag{2.11}$$

An important property of the affine Hilbert function is the following:

Theorem 2.1.6. Let $I \subset \mathbb{F}[X_1, \ldots, X_n]$ be an ideal, then

$$^a\mathrm{HF}_I(s) = {^a\mathrm{HF}}_{\mathrm{LT}(I)}(s),\ s \geq 0. \tag{2.12}$$

Another main result is the existence of the Hilbert polynomial.

Theorem 2.1.7. There exist a polynomial $^a\mathrm{HP}_I(t) \in \mathbb{Q}[t]$ such that

$$^a\mathrm{HP}_I(s) = {^a\mathrm{HF}}_I(s),\ \forall s \geq 0. \tag{2.13}$$

Proof (sketch):

- Find a Groebner basis G of I
- Let $M \subset \mathbb{Z}_{\geq 0^n}$ be such that $\{X^\beta\}_{\beta \in M} = \{\mathrm{LM}(g) : g \in G\}$
- Write

  $$C(I) = \bigcap_{\beta \in M} C(\beta), C(\beta) = \{\alpha | X^\beta \nmid X^\alpha\} \tag{2.14}$$

  as a union of $C(J, \tau) = \{\alpha | \alpha_j = \tau(j)\ \forall j \in J\}$. To do this use

  $$C(\beta) = \bigcup_{i=1}^{n} \bigcup_{t_i=0}^{\beta_i - 1} C(\{i\},\ \tau : i \mapsto t_i). \tag{2.15}$$

- Use the formula $|C(J,\tau)_{\leq s}| = \binom{d+s-|\tau|}{d}, d = n - |J|, |\tau| = \sum_{j\in J}\tau(j)$ and the inclusion-exclusion principle to compute

$$
{}^a\mathrm{HP}_I(s) = |C(I)_{\leq s}| = \left|\bigcup_{(J,\tau)} C(J,\tau)_{\leq s}\right|. \tag{2.16}
$$

Example 2.1.8. Let $I = (X_1 - X_3^2, X_2 - X_3^3)$ and fix the LEX order with $X_1 > X_2 > X_3$. Then we calculate

- $G = \{X_1 - X_3^2, X_2 - X_3^3\}$
- $M = \{(1,0,0),(0,1,0)\}$
- We have

$$
\begin{aligned}
C(I) &= C(\{1\}, \tau : 1 \mapsto 0) \cap C(\{2\}, \tau : 2 \mapsto 0) \\
&= C(\{1,2\}, \tau : 1 \mapsto 0, 2 \mapsto 0).
\end{aligned} \tag{2.17}
$$

So we conclude that $|C(I)|_{\leq s} = \binom{1+s-0}{1} = s+1$.

- (Algebraic dimension). Here we define the dimension of an algebraic set.

Definition 2.1.9. Let $V \subset \mathbb{F}^n$ be an algebraic set. We define its dimension as $\dim(V) = \deg {}^a\mathrm{HP}_{I(V)}(t)$. It has the property that it's the biggest integer $d \geq 1$ such that there exist indexes $1 \leq i_1 < \cdots < i_d \leq n$ such that $I(V) \cap \mathbb{F}[X_{i_1}, \ldots, X_{i_d}] = (0)$.

Example 2.1.10. Let $H_f = V(f)$, f irreducible, then

$$
\begin{aligned}
\dim(H_f) &= \deg {}^a\mathrm{HP}_{I(H_f)}(t) \\
&= \deg {}^a\mathrm{HP}_{(f)}(t) \\
&= \deg(\frac{\deg(f)}{(n-1)!}t^{n-1} + \cdots) \\
&= n-1.
\end{aligned} \tag{2.18}
$$

The dimension has the expected property: If $W \subset V$ is closed, then $\dim(W) \leq \dim(V)$. In fact

$$
\begin{aligned}
&I(V) \subset I(W) \Rightarrow \\
&I(V)_{\leq s} \subset I(W)_{\leq s} \Rightarrow \\
&{}^a\mathrm{HF}_{I(W)}(s) \leq {}^a\mathrm{HF}_{I(V)}(s),\ \forall s \geq 0 \Rightarrow \\
&\deg {}^a\mathrm{HF}_{I(W)}(t) \leq \deg {}^a\mathrm{HF}_{I(V)}(t).
\end{aligned} \tag{2.19}
$$

- (Transcendence degree). Let A be a field which is also a finitely generated $\mathbb{F}$-algebra. We define the transcendence degree of A over $\mathbb{F}$ (and write $\partial_{\mathbb{F}}(A)$) to be largest size of an algebraicaly independent subset of A over $\mathbb{F}$.

Example 2.1.11. We have $\partial_{\mathbb{F}}(\mathbb{F}[X_1,\dots,X_n]) = n$. Indeed, let $\{f_1,\dots,f_r\}$ be polynomials and suppose that $r > n$. Write

$$\begin{aligned} &W_s = \{f^\beta : \beta \in \mathbb{Z}^r_{\geq 0},\ |\beta| \leq s\},\ d = \max\{\deg(f_i) : i = 1,\dots,r\} \\ &V_s = \{X^\alpha : |\alpha| \leq s\} \end{aligned} \tag{2.20}$$

Then $W_s \subset V_{ds}$ and

$$|W_s| = \binom{r+s}{s} \overset{s \gg 0}{>} \binom{n+sd}{sd} = \dim(V_{sd}) \geq \dim(W_s). \tag{2.21}$$

So, for $s \gg 0$ there exist $\{a_\beta\}_\beta \subset \mathbb{F}$ such that $\sum_\beta a_\beta f^\beta = 0$ and $\{f_1,\dots,f_r\}$ is algebraicaly dependent.

Theorem 2.1.12. Let V be an algebraic set and write $I = I(V)$, then

$$\dim(V) = \partial_{\mathbb{F}}(A[V]) = \partial_{\mathbb{F}}(\frac{\mathbb{F}[X_1,\dots,X_n]}{I}). \tag{2.22}$$

Proof: Let $d = \dim(V)$, so there exist indexes $1 \leq i_1 < \cdots < i_d \leq n$ such that $I \cap \mathbb{F}[X_{i_1},\dots,X_{i_d}] = (0)$. This is equivalent to say that $[X_{i_1}],\dots,[X_{i_d}] \in A[V]$ are algebraicaly independent over $\mathbb{F}$. Therefore $d \leq \partial_{\mathbb{F}}(A[V])$. Now, suppose that $\varphi_i = [f_i] \in A[V], i = 1,\dots m$ are algebraicaly independent over $\mathbb{F}$ and put $N = \max\deg(f_i)$, then the map

$$\mathbb{F}[Y_1,\dots,Y_m]_{\leq s} \mapsto \frac{\mathbb{F}[X_1,\dots,X_n]_{\leq Ns}}{I_{\leq Ns}} : g \mapsto g(\varphi_1,\dots,\varphi_m) \tag{2.23}$$

is injective. Then, by taking dimensions we get

$$\binom{m+s}{s} \leq {}^a\mathrm{HF}_I(s),\ \forall s \geq 0. \tag{2.24}$$

We know that both expressions are polynomials in s, so by taking degrees we conclude that $m \leq \dim(V) = d$ and $\partial_{\mathbb{F}}(A[V]) \leq d$. $\square$

With small changes in the above proof, we can verify that in the case of $A[V]$ being an integral domain, we have also $\partial_{\mathbb{F}}(K) = \dim(V),\ K = \mathrm{Frac}(A[V])$.

Example 2.1.13. Let $C = V(X_1 - X_3^2, X_1 - X_3^3)$, then $I(C) = (X_1 - X_3^2, X_2 - X_3^3)$ and

$$\frac{\mathbb{F}[X_1,X_2,X_3]}{I(C)} \cong \mathbb{F}[X_3] \tag{2.25}$$

so $\partial_{\mathbb{F}}(A[C]) = 1$.

Example 2.1.14. The purpose of this example is to show that if $V \subset \mathbb{F}^n, |\mathbb{F}| = \infty$ is a linear space, then $\dim(V) = \dim_{\mathbb{F}}(V)$.

– Let $\mathrm{Ann}(V) = \mathrm{Span}(f_1, \ldots, f_m)$. Since V has finite dimension, we have $v \in V \iff f(v) = 0\, \forall f \in \mathrm{Ann}(V)$ and $V = V(\{f_1, \ldots, f_m\})$.

– Let $f_i = \sum a_{ij} X_j$, where we identify X_j with the jth coordinate functions in $\mathbb{F}^n$. Now, if $A = (a_{ij})$ and apply reduction to put A in the form:

$$\begin{pmatrix} a_{1j_1} & * & 0 & * & \cdots & 0 & \cdots \\ & & a_{2j_2} & * & \cdots & 0 & \cdots \\ & & & & \vdots & \cdots & \vdots \\ & & & 0 & \cdots & a_{mj_m} & \cdots \end{pmatrix} \tag{2.26}$$

with $a_{ij_i} = 1,\ j_1 < j_2 < \cdots$. Now put LEX with: $X_{j_1} > \cdots X_{j_m} > X_j\ j \notin \{j_1, \ldots, j_m\}$ and write $f_i = X_{j_i} + h_i$ with $h_i \in \mathbb{F}[X_j : j \notin \{j_1, \ldots, j_m\}]$. Then we see that $A[V] \cong \mathbb{F}[X_j : j \notin \{j_1, \ldots, j_m\}]$ and $(f_1, \ldots, f_m)$ is prime. Therefore $I(V) = (f_1, \ldots, f_m)$. Finally

$$\dim(V) = \partial_{\mathbb{F}}(A[V]) = \dim_{\mathbb{F}} \mathbb{F}[X_j : j \notin \{j_1, \ldots, j_m\}] = \dim_{\mathbb{F}} V. \tag{2.27}$$

2.2 Tangent Spaces

- (Regular maps). Let $V \subset \mathbb{F}^m, W \subset \mathbb{F}^n$ be algebraic sets. Then a map $f : V \to W$ is said to be regular if there exist polynomials such that $f = (f_1, \ldots, f_m)$. Such a map is continuous in the Zariski topology.

  Remark 2.2.1. We can identify $A[V]$ with the set of regular functions $V \to \mathbb{F}$. In this context, we have a contravariant functor:

  $$\varphi : V \to W \rightsquigarrow \varphi^* : A[W] \to A[V] : [g] \mapsto [g] \circ \varphi \tag{2.28}$$

  Example 2.2.2. Let $\varphi : \mathbb{F} \to C : x \mapsto (x^3, x^2, x)$ and write $C = (X_1 - X_3^2, X_2 - X_3^3)$. Remember the isomorphism $A[C] \to \mathbb{F}[Y] : [X_1] \mapsto Y^2, [X_2] \mapsto Y^3, [X_3] \mapsto Y$. So we have

  $$\varphi^* : A[C] \to A[\mathbb{F}] : [X_3] \mapsto [X_3] \circ \varphi. \tag{2.29}$$

  The first map is $[X_3] : C \to \mathbb{F} : (x^3, x^2, x) \mapsto x$ and the last map is $\mathbb{F} \to C : x \mapsto x$. At the algebraic level, the map φ^* is just the the isomorphism $\mathbb{F}[Y] \cong \mathbb{F}[X]$.

- (Tangent spaces). Let V be an algebraic set and $V_p = \mathbb{F}$ be the $A[V]-$module with action $[f]\alpha \equiv f(p)\alpha$. Let $I(V) = (f_1, \ldots, f_m)$. We define

  $$\begin{aligned} T_p(V) &= \{v \in \mathbb{F}^n : d(f)_p(v) = 0\ \forall f \in I(V)\} \\ &= \{v \in \mathbb{F}^n : d(f_i)_p(v) = 0\ \forall i\} \\ &= V(\{d(f_i)_p : i = 1, \ldots, m\}) \end{aligned} \tag{2.30}$$

Remark 2.2.3. Caution ! The polynomials of f_i here are not the generators of V, but of $I(V)$. For example, if $V = V(f)$, we can conlude that $T_p(V) = V(df_p)$ only if f is irreducible, i.e., if $I(V) = (f)$.

Here $df_p \equiv \sum_{i=1}^n D_i(f)(p)X_i \in \mathbb{F}[X_1, \ldots, X_n]$. We have an isomorphism

$$T_p(V) \to \mathrm{Der}_{\mathbb{F}}(A[V], V_p) : v \mapsto D_v;\ D_v([f]) = \sum_{i=1}^n [D_i(f)] \cdot v_i \in \mathbb{F}. \tag{2.31}$$

with inverse $D \mapsto (D([X_i]))_{i=1}^n$. Of course $D_v([f]) = df_p(v)$.

Maybe it will be better to change notation of V_p to ${}^p\mathbb{F}$ or something.

Example 2.2.4. Let $C = (f_1, f_2),\ f_1 = X_1 - X_3^2, f_2 = X_2 - X_3^3$ and $p = (x^3, x^2, x) \in C$. We have

$$d(f_1)_p = 1 - 2xX_3,\ d(f_2) = 1 - 3x^2X_3 \tag{2.32}$$

and $T_p(C) = \mathbb{F}\langle(3x^2, 2x, 1)\rangle$.

Example 2.2.5. The determinant is an irreducible polynomial. Indeed, let $\det \in \mathbb{F}[X_{ij} : 1 \le i, j \le n]$. Then, if we apply Laplace theorem on the first row, we get

$$\det = A_{11}X_{11} - A_{12}X_{12} + \cdots + (-1)^n A_{1n}X_n \tag{2.33}$$

where A_{ij} denotes the determinant of the matrix obtained by excluding the ith row and jth collumn. By induction on the number of variables, we assume that A_{ij} are irreducible polynomials. Now, suppose that $\det = fg$, where $\deg(f), \deg(g) > 0$. Choose LEX with $X_{ij} > X_{k\ell}$ if $i < k$ or if $i = k$ and $j < \ell$. Then

$$\mathrm{LT}(A_{11})X_{11} = \mathrm{LT}(\det) = \mathrm{LT}(f)\mathrm{LT}(g). \tag{2.34}$$

Assume that $X_{11}|\mathrm{LT}(f)$ (so $\mathrm{LT}(g) < X_{11}$) and write $f = AX_{11} + B,\ \mathrm{LT}(B) < X_{11}$. Then we must have $AgX_{11} = A_{11}X_{11}$ and $A_{11} = Ag$. Since $\deg(g) > 1$ and A_{11} is irreducible, we must have $g = A_{11}$ and $A = 1$. Now we get

$$-A_{12}X_{12} + \cdots + (-1)^n A_{1n}X_n = BA_{11} \tag{2.35}$$

and $-\mathrm{LT}(A_{12})X_{12} = \mathrm{LT}(B)\mathrm{LT}(A_{11})$. Since $X_{22}|\mathrm{LT}(A_{11})$, we are led to $X_{22}|\mathrm{LT}(A_{12})$ which is a contradiction because X_{22} doenst appear in A_{12} by definition.

Now, since det is an irreducible homogeneous polynomial, the difference $\det -1$ must be irreducible. Therefore $I(SL_n\mathbb{F}) = (\det -1)$ and we can compute the tangent space

$T_I(SL_n) = V(d_I(\det))$. We have

$$\begin{aligned} d_I \det &= \sum_{k,\ell} D_{k\ell}(\det)(I)X_{k\ell} \\ &= \sum_{k,\ell}\sum_{\sigma\in S_n} D_{k\ell}\prod_{i=1}^{n} X_{i\sigma(i)}(I)X_{k\ell} \\ &= \sum_{k,\ell}\sum_{\sigma\in S_n} \delta_{\ell,\sigma(k)}\prod_{i\neq k}\delta_{i,\sigma(i)}X_{k\ell} \\ &= \sum_{k,\ell}\delta_{k\ell}X_{k\ell} \quad (\delta_{\ell,\sigma(k)}\prod_{i\neq k}\delta_{i,\sigma(i)} \neq 0 \text{ only if } \sigma = 1) \\ &= \sum_{k} X_{kk}. \end{aligned} \tag{2.36}$$

Therefore $T_I SL_n(\mathbb{F}) = \mathfrak{sl}_n\mathbb{F}$, the Lie algebra of traceless matrices.

- (Gauss lemma). We state the Gauss lemma.

Lemma 2.2.6. Let R be a UFD ring (think in $\mathbb{F}[X]$), then $R[X]$ is a UFD.

Proof: Let f and $K = \mathrm{Frac}(R)$. Since $K[X]$ is a UFD, we have $f = FG,\ F, G \in K[X]$ with F irreducible and $\deg(G) < \deg(f)$. Since R is a UFD, there exist $r, s \in R$ such that $rF, sG \in R[X]$. Decompose $rs = t_1 \cdots t_k$ into irreducible elements. Then if $F' = rF, G' = sG$, we have

$$[F'][G'] = [rs][f] = 0 \in R/(t_i)[X] \tag{2.37}$$

because $f \in R[X]$. So, since $R/(t_i)[X]$ is a domain, we see that $t_i|F'$ or $t_i|G'$. Then we see that there exist a partition $\{1,\ldots,k\} = I \cup J$ such that $F'/\prod_{i\in I} t_i, G'/\prod_{j\in J} t_j \in R[X]$. It's clear that F' still irreducible, then an induction on degree yields the result. $\square$

Corollary 2.2.7. Let $f, g \in R = \mathbb{F}[X_1,\ldots,X_n]$ and assume f irreducible with $f \nmid g$, then there exist $F, G \in R$ such that

$$Gf + Fg = h \in \mathbb{F}[X_1,\ldots,X_{n-1}] \setminus 0. \tag{2.38}$$

Proof: Let $K = \mathrm{Frac}(\mathbb{F}[X_1,\ldots,X_{n-1}])$, then $f, g \in K[X_n]$ and there exist $A, B \in K[X_n]$ such that $Af + Bg = 1$ (by Gauss lemma f still irreducible as an element of $K[X_n]$). Then there exist $h \in \mathbb{F}[X_1,\ldots,X_{n-1}]$ such that $hA, hB \in \mathbb{F}[X_1,\ldots,X_{n-1}]$. So by writing $G = hA, F = hB$ we are done. $\square$

Notation 2.2.8. Let $A = \mathbb{F}[X_1,\ldots,X_n]/I$, where M is a $A-$module. We write

$$\mathcal{T}_{A,M} = \{(v_1,\ldots,v_n) \in M^n : \sum_i [D_i(f)]\cdot v_i,\ \forall f \in I\}. \tag{2.39}$$

We have the isomorphism $\mathcal{T}_{A,M} \to \mathrm{Der}_{\mathbb{F}}(A,M) : v \mapsto D_v$ with inverse

$$D \mapsto (D(\overline{X_i}))_{i=1}^n. \tag{2.40}$$

Now consider the following context, let $K = \mathrm{Frac}(A)$ as A-module (A acts by multiplication). Then we have an isomorphism

$$\mathrm{Der}_{\mathbb{F}}(K,K) \to \mathrm{Der}_{\mathbb{F}}(A,K) : D \mapsto D|_A. \tag{2.41}$$

In particular we also have the isomorphism $\mathcal{T}_{A,K} \cong \mathrm{Der}_{\mathbb{F}}(K,K)$. We observe that this is also a $K-$vector space isomorphism, it's a bijection compatible with the $K-$action. From this isomorphism, we see that $\mathcal{T}_{A,K}$ only depends on K, and the fact that $\mathrm{Frac}(A) = K$.

Theorem 2.2.9. Let $A = \mathbb{F}[X_1, \ldots, X_n]/I$ be a domain and $K = \mathrm{Frac}(A)$. Then

$$\partial_{\mathbb{F}}(A) = \dim_K \mathrm{Der}_{\mathbb{F}}(K,K). \tag{2.42}$$

Proof: Let $d = \partial_{\mathbb{F}}(A) = \partial_{\mathbb{F}}(K)$, then there exist $z_1, \ldots, z_{d+1}$ such that

- $K = \mathbb{F}(z_1, \ldots, z_{d+1})$
- $\{z_1, \ldots, z_d\}$ is algebraicaly independent
- z_d is separable over $\mathbb{F}(z_1, \ldots, z_d)$

Let $R = \mathbb{F}[z_1, \ldots, z_{d+1}] \subset K$ and $f \neq 0$ irreducible such that $f(z_1, \ldots, z_{d+1}) = 0$. Then by corollary (2.2.7), we see that $\alpha : K[Y_1, \ldots, Y_{d+1}] \to K : g \mapsto g(z_1, \ldots, z_{d+1})$ has kernel (f). Therefore

$$\begin{aligned}\mathcal{T}_{R,K} &= \{(v_1, \ldots, v_{d+1}) \in K^{d+1} : \sum_i [D_i(f)] \cdot v_i = 0\} \\ &= \{(v_1, \ldots, v_{d+1}) \in K^{d+1} : \sum_i D_i(f)(z_1, \ldots, z_{d+1}) \cdot v_i = 0\}.\end{aligned} \tag{2.43}$$

We see that $D_i(f)(z_1, \ldots, z_{d+1}) = 0 \Rightarrow D_i(f) \in (f)$ and $D_i(f) = 0$. Since $f \neq 0$ and $|\mathbb{F}| = \infty$, we must have $D_i(f) \neq 0$ for some i, so $D_i(f)(z_1, \ldots, z_{d+1}) \neq 0$ for some i. Then we conclude that

$$\dim_K \mathrm{Der}_{\mathbb{F}}(K,K) = \dim_K \mathcal{T}_{R,K} = d = \partial_{\mathbb{F}}(K) = \partial_{\mathbb{F}}(A). \tag{2.44}$$

$\square$

Corollary 2.2.10. If $A = \mathbb{F}[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$, then

$$\partial_{\mathbb{F}}(A) = \dim_K \mathcal{T}_{A,K} = n - \mathrm{Rank}_K([D_i(f_j)]). \tag{2.45}$$

Example 2.2.11. Let $C = V(f_1, f_2), f_1 = X_1 - X_3^2, f_2 = X_2 - X_3^3$. We have

$$A[C] = \frac{\mathbb{F}[X_1, X_2, X_3]}{(f_1, f_2)} \to \mathbb{F}[X_3] : [g] \mapsto g(X_3^3, X_3^2, X_3). \tag{2.46}$$

So $A[C]$ is a domain and we can define $K = \mathrm{Frac}(A[C])$. We compute

$$\mathrm{Rank}_K([D_i(f_j)]) = \mathrm{Rank}_K \begin{pmatrix} 1 & 0 & -2[X_3] \\ 0 & 1 & -3[X_3] \end{pmatrix} = 2. \tag{2.47}$$

Then we conclude $\dim(V) = 3 - 2 = 1$.

- (differential of a regular map). Here we define the notion of differential of a regular map.

  Definition 2.2.12. Let $\varphi : V \to W$ be a regular map between algebraic varieties. Also, let $p \in V, q = \varphi(p) \in W$. We define:

  $$d\varphi_p : \mathrm{Der}_{\mathbb{F}}(A[V], V_p) \to \mathrm{Der}_{\mathbb{F}}(A[W], W_q) : D_v \mapsto D_v \circ \varphi^* \tag{2.48}$$

  At the level of tangent spaces, we have:

  $$\begin{tikzcd}
  \mathrm{Der}_{\mathbb{F}}(A[V], V_p) \arrow[rr, "d\varphi_p"] & & \mathrm{Der}_{\mathbb{F}}(A[W], W_q) \arrow[dd] & \\
  & D_v \arrow[rr, mapsto] & & D_v \circ \varphi^* \arrow[dd, mapsto] \\
  T_p(V) \arrow[uu] \arrow[rr, dashed] & & T_q(W) & \\
  & v \arrow[uu, mapsto] \arrow[rr, mapsto] & & (D_v \circ \varphi^*([Y_i]))_{i=1}^m
  \end{tikzcd}$$

  In more details, if we write $\varphi = (f_1, \dots, f_m)$, then $\varphi^*([Y_i]) = [f_i]$ and we get

  $$D_v([f_i]) = \sum_{j=1}^m [D_j(f_i)] \cdot v_j = \sum_{j=1}^m D_j(f_i)(p) v_j. \tag{2.49}$$

  Therefore

  $$d\varphi_p(v) = (D_j(f_i)(p))_{i,j} \cdot v = J_p(\varphi) \cdot v. \tag{2.50}$$

  Example 2.2.13. Let $G \subset \mathbb{F}^{n^2}$ be an algebraic group and $\mu : G \times G \to G$ be the product of matrices. We have $G \times G \subset \mathbb{F}^{n^2} \times \mathbb{F}^{n^2} \cong \mathbb{F}^{2n^2}$. Now, identify the first n^2

coordinate functions with X_{ij} and the second n^2 coordinate functions with Y_{ij}. So we have

$$\mu_{ij} = \sum_{\ell} X_{i\ell} Y_{\ell j}. \tag{2.51}$$

Then we have

$$\frac{\partial \mu_{ij}}{\partial X_{rs}}(I) = \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) = \delta_{ir}\delta_{sj}. \tag{2.52}$$

Therefore

$$d_{(I,I)}\mu(A,B) = (\sum_{r,s} \frac{\partial \mu_{ij}}{\partial X_{rs}}(I) A_{rs} + \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) B_{rs})_{ij} = (A_{ij} + B_{ij}) = A + B. \tag{2.53}$$

TODO:

1. Prove that $T_1 G$ is a Lie algebra. Remember the star operator.
2. Compute the Lie algebra of SL_n using the fact that $\det -1$ is an irreducible polynomial.

Chapter 3

List of exercises

Exercise 1. Let $U = k^2 - \{0\}$, show that the set of regular functions $U \to k$ are polynomials, then conclude that U is not an affine variety.

Solution. Let $\varphi \in \mathcal{O}_U(U)$ be a regular function, then $\varphi|_{D_{k^2}(X)} = f/X^m$ and $\varphi|_{D_{k^2}(Y)} = g/Y^n$ and on the intersection $D_{k^2}(X) \cap D_{k^2}(Y) = D_{k^2}(XY)$, we have

$$Y^n f - X^m g = 0 \quad (*). \tag{3.1}$$

Since there exist an infinity number of points $p \in D_{k^2}(XY)$ (assuming k is algebraically closed), we also have the equality $(*)$ in $k[X,Y]$. Now, let $f = \sum a_{ij} X^i Y^j$ and $g = \sum b_{ij} X^i Y^j$. Then, equality $(*)$ means

$$a_{i,j-n} = a_{i-m,j}, \ \forall i \geq m, j \geq n. \tag{3.2}$$

Therefore $f/X^m = g/Y^n \in k[X,Y]$ and φ is a polynomial, because $U = D_{k^2}(X) \cup D_{k^2}(Y)$. In conclusion, we proved that the inclusion $i : U \hookrightarrow k^2$ induces a ring isomorphism

$$i^{\#}(k^2) : k[X,Y] \to \mathcal{O}_U(U) : f \mapsto f|_U \ (**). \tag{3.3}$$

Now assume $U = V(S), \ S \subset k[X,Y]$, then $f \in S \Rightarrow f|_U = 0$ and $f = 0$ because of $(**)$. So $S = \{0\}$ and $U = k^2$, which is a contradiction.

Furthermore, if U is isomorphic to an affine algebraic set, then we would have an injective morphism $\varphi : V(S) \subset k^2 \hookrightarrow k^2$ (which is not necesseraly an inclusion) such that the induced map

$$\varphi^{\#} : k[X,Y] \to \mathcal{O}_V(V) \cong k[X,Y]/I(V) \tag{3.4}$$

is an isomorphism. However, if $h \in k[X,Y]$ and $h|_V = 0$, then $(\varphi^{\#})^{-1}(\overline{h}) = 0$. This means that $h(F,G) = 0$, where $F = (\varphi^{\#})^{-1}(\overline{X}), \ G = (\varphi^{\#})^{-1}(\overline{Y})$.

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